

MATB24H3

Exercise Questions

Student Name: Your Name

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2.1.1. Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ a binary operation?

Solution:

2.1.2. Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? what is the identity element? How about $(\mathcal{P}(\mathbb{N}), \cup)$ for any arbitrary set X ?

Solution:

2.2.1. Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}), +, \cdot)$ forms a field. What happens if $d = m^2$ for some m ?

Solution:

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

Solution:

2.2.3. Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and the additive inverse element $-a$ of any element $a \in F$ are unique.

Solution:

2.2.4. Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

Solution:

2.2.5. Prove that the triple $(\mathbb{Z}_p, + \bmod p, \cdot \bmod p)$ is a field $\iff p$ is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

Solution:

2.2.6. Prove that the 2-tuple $(S, \circ)$, where S is the set of symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

Solution:

2.2.7. Characterise all idempotent elements of $(\mathcal{M}_{2 \times 2}(\mathbb{R}), +, \cdot)$.

Solution:

2.2.8. Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

Solution:

2.2.9. Let $(\mathcal{R}, +, \cdot)$ be a ring and suppose that $\alpha \in \mathcal{R}$ has a multiplicative inverse. Moreover, suppose that $\alpha \in \mathcal{C}(\mathcal{R})$. Is $\alpha^{-1} \in \mathcal{C}(\mathcal{R})$?

Solution:

2.2.10. Does the set of functions with infinitely many roots form a subring of $\mathcal{F}(\mathbb{R})$? Justify your answer.

Solution:

2.2.11. Prove the following statements:

- (a) There exist rings $(\mathcal{R}, +, \cdot)$ with an identity element $1_{\mathcal{R}}$ such that $1_{\mathcal{R}} \notin S$ for some subring S .
- (b) [TERM TEST QUESTION] There may exist a ring $\mathcal{R}$ with an identity element $1_{\mathcal{R}}$ and a subring $S \subseteq \mathcal{R}$ such that S has its own identity element 1_S , where $1_S \neq 1_{\mathcal{R}}$.

Solution:

2.2.12 Determine the zero and the identity elements of the polynomial ring $(\mathcal{R}[x], +, \cdot)$ where $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2 \times 2}(\mathbb{R}))$.

Solution:

2.2.13 True/False. Justify:

1. Let $\mathbb{F}$ be any field. Suppose that α and β are two elements of an extension field $\mathbb{K}$ of $\mathbb{F}$. Then $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are always distinct extension fields of $\mathbb{F}$.
2. Given any set X , there is no operation $\cdot$ that can be defined on $[X]^{\leq \|X\|}$ that makes it a group.

3. Given a monoid $(\mathcal{S}, *)$ there exists a binary operation $\cdot$ on $\mathcal{S}$ such that $(\mathcal{S}, \cdot)$ no longer admits an identity element. If so, what assumption did you need to make?
4. $(\mathbb{R}, \cdot)$ forms a group.
5. $(\mathbb{R}, +)$ forms a group with the identity element being 1.
6. Suppose $(\mathbb{F}, +, \cdot)$ is the field. Then $(\mathbb{F}, +)$ and $(\mathbb{F}, \cdot)$ both form a group.

Solution: